

# Dr Lawrence Yolland

## SOFTWARE ENGINEER

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Researcher-turned-software engineer with a multidisciplinary background spanning computational biology, data analysis, and full-stack development. Over six years of experience delivering high-quality, client-facing React applications and building with Node.js, with prior experience developing custom image-processing tools in Python and MATLAB. Equally comfortable working with research scientists, product teams, and business stakeholders. Motivated by problems at the intersection of rigorous analysis and practical engineering.

## EXPERIENCE

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### Software Engineer

Red Bee Media | 2020 - current

- **Internal release-tracking tool:** Technical owner of a full-stack TypeScript internal tool for deployment managers and engineers to track releases and view component metadata. Designed and built out the Node.js/Express backend, integrating third-party APIs across Jenkins and AWS (S3, Cognito, STS, DynamoDB, QuickSight), with real-time WebSocket updates and custom D3 visualisations for the React frontend. Subsequently rewrote the backend in Go, aligning with the wider engineering team's preferred stack.
- **Workflow monitoring dashboard:** Lead developer of a customer-facing React dashboard with fully customisable tiles displaying workflow summary statistics and detailed graphical renderings.
- **Broadcast payout dashboard:** Lead developer on a business-critical broadcast payout dashboard used by operations teams across major broadcasters. Scoped and delivered a plan of work that included a security audit, dependency updates, codebase modernisation, and alignment with our established technical stack.
- **Solutions Architect tooling:** Built two React applications for Solutions Architects, translating Figma designs into production-ready interfaces with interactive D3 visualisations and a custom JSON editor built with CodeMirror.
- **Standards & best practices:** Championed TypeScript and Redux Toolkit adoption across applications; authored business cases and impact assessments for technology decisions including an Axios-to-native-Fetch migration and updated security standards.
- **Team & delivery:** Worked across the full application suite requiring both depth and breadth of codebase knowledge: triaging bugs, developing features, reviewing code, and producing documentation in an agile workflow. Mentored new hires through onboarding, setting SMART goals and co-developing training pathways.

### Data Analyst

Contentsquare | 2018 - 2019

- **SQL & data analysis:** Analysed large-scale web analytics datasets using SQL, Python, and Contentsquare's proprietary platform to surface actionable UX insights for enterprise clients.
- **Automation & 3rd-party integration:** Identified a manual extraction bottleneck and independently automated it in Python on a cron schedule, integrating with the Google Sheets and Data Studio APIs to deliver client dashboards before the working day began.
- **Communication:** Led cross-functional workshops translating complex data findings into concrete product recommendations for developers, product managers, and UI specialists.

### PhD Researcher

University College London (BBSRC London Interdisciplinary Doctoral Programme) | 2014 - 2018

- **Computational research:** Developed novel computational image-processing tools in MATLAB and Python to analyse immune cell migration.
- **Collaboration:** Part of a multidisciplinary team of engineers, geneticists, and cell biologists. Contributed methods and tooling to further publications in Nature Communications, Current Biology, and the Journal of Cell Science.
- **Publications & Presentations:** Thesis work published in Nature Cell Biology (2019). Spoke at the Abercrombie Meeting (Royal Microscopical Society, held every five years). Won Best Talk at the British Actin Meeting.

## CURRENTLY WORKING ON

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**Project Scout:** A tool that gives junior frontend developers senior-level feedback on their portfolio projects before they put them in front of a technical reviewer. Junior developers rarely get professional code review until they're already in the hiring process. Project Scout will surface the flags that would concern a technical reviewer - exposed credentials, missing or conflicting lockfiles, high complexity and no test coverage, and structural red flags - weighted by severity and passed to an LLM to generate actionable recommendations. A solo full-stack project - a Hono API layer handles repo submission and cloning, a Java static analyser produces the signal report, and React/Astro on the frontend.

## SKILLS

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### Languages

TypeScript / JavaScript, Python, Go, Ruby, Java, MATLAB

### Frontend

React, HTML / CSS, SCSS, Tailwind CSS, Redux

### Backend

Node.js, Express, Ruby on Rails, RESTful APIs, WebSockets, SQL, NoSQL

### Tools & Platforms

Git (GitLab / GitHub), AWS (studying for SAA-C03), Jenkins, Jira, Confluence, Figma, Jest, Vite

### General Skills

Project management, technical communication, data analysis, data visualisation, performance optimisation, research

## EDUCATION

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**University College London** 2014 - 2018 BBSRC London Interdisciplinary Doctoral Programme. Interdisciplinary research alongside engineers, cell biologists, and material scientists. Thesis work published in Nature Cell Biology.

**University of Kent** 2013 - 2014 MSc Physics (Research): Astrobiology research project published in Philosophical Transactions of the Royal Society A.

**University of Kent** 2010 - 2013 BSc Hons Biomedical Sciences, 2:1: Modules included Immunology, Genetics, and Statistics.

## ADDITIONAL TRAINING

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**Flatiron School** 2019 Full Stack Software Engineering (HTML, CSS, JavaScript, Ruby; React + Ruby on Rails capstone)

## OUTSIDE INTERESTS

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Science, technology, and entrepreneurship, with a focus on the intersection of research and innovation. Co-hosted and edited the Radical Science Podcast, interviewing founders, open science contributors, and metascience researchers. Built a newsletter reaching ~2,500 subscribers and managed a community of ~800 scientists, technologists, and entrepreneurs. Organised events spanning biotech, future space, quantum computing and more, bringing together startups, academics, and industry practitioners. Served for several years as an Innovation Jury Member for the SXSW Innovation Awards. Outside of work, I enjoy trail running, swimming, strength training, Jiu Jitsu, and gravel cycling.

## PUBLICATIONS

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**AL Law et al.** Nance-Horan Syndrome-like 1 protein negatively regulates Scar/WAVE-Arp2/3 activity and inhibits lamellipodia stability and cell migration. Nature Communications, 2021.

**L Yolland et al.** Persistent and polarized global actin flow is essential for directionality during cell migration. Nature Cell Biology, 2019.

**S Brayford et al.** Heterotypic contact inhibition of locomotion can drive cell sorting between epithelial and mesenchymal cell populations. Journal of Cell Science, 2019.

**BJ Sanchez-Sanchez et al.** A moving source of matrix components is essential for de novo basement membrane formation. Current Biology, 2017.

**MJ Burchell, KH McDermott, MC Price, LJ Yolland.** Survival of fossils under extreme shocks induced by hypervelocity impacts. Philosophical Transactions of the Royal Society A, 2014.